Refined Humbert invariants and isogeny problems

Isogeny Club, Season 8

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Abelian varieties

Theory of quadratic forms

Refined Humbert invariants

Applications to isogeny-based cryptography

Abelian varieties

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Let $\varphi : A_1 \to A_2$ be a homomorphism of abelian varieties. If φ is surjective and has finite kernel, then φ is called an *isogeny*.

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- Let (A, θ) and $(A', \theta^{'})$ be p.p.a.v. A morphism $\varphi : A \to A^{'}$ is said to be an isogeny of p.p.a.v if there exists an integer $m \geq 1$ such that $\hat{\varphi} \circ \theta^{'} \circ \varphi = [m]\theta$.

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- The refined Humbert invariant $q_{(A,\theta)}$ is a *positive definite integral quadratic form* attached to a principally polarized abelian surface (A, θ) [Kani, 94].
- Many geometric properties of (A, θ) are translated into arithmetic properties of $q_{(A,\theta)}$.

Theory of quadratic forms

Binary, ternary, and quintic quadratic forms

Definition

Let R be a commutative ring with 1. A *quadratic form* over R in n variables is polynomial $f(x_1, .., x_n)$ of the form

$$f = f(x_1, ..., x_n) = \sum a_{ij} x_i x_j \text{ where } a_{ij} \in R.$$

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- An integral ternary quadratic form

  $$f(x, y, z) = ax^2 + by^2 + cz^2 + ryz + sxz + txy$$

  abbreviated by

  $$[a, b, c, r, s, t].$$

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- f is called *positive definite* if it only represents (for $x \neq 0$) positive integers.
- Let f_1,f_2 be n-ary quadratic forms over $\mathbb{Z}$. We say that two forms f_1 and f_2 are *equivalent*, if we have $f_2 = f_1 M$, for some $M \in \mathrm{GL}_n(\mathbb{Z})$.

- For a unique symmetric matrix A, f can be expressed as $f(x) = \frac{1}{2}x^t Ax$. The matrix A is the *coefficient matrix* of f, and $det(f) = det(A)$.

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- Two equivalent integral quadratic forms have the same discriminant, primitively represent the same integers.

Refined Humbert invariants

The quadratic forms q_A and q_{E_1,E_2}

Definition

The intersection product $(D_1 \cdot D_2)$ of divisors D_1, D_2 on an abelian surface A defines an indefinite integral quadratic form q_A on $\mathrm{NS}(A)$, called *intersection form*

$$q_A(D) = \frac{1}{2}(D \cdot D) \text{ for all } D \in \mathrm{Div}(A).$$

in ρ variables, where ρ is the Picard number of A.

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The *degree map* is a positive definitive integral quadratic form q_{E_1,E_2} such that

$$q_{E_1,E_2}(\varphi) = \deg(\varphi) \text{ for any } \varphi \in \mathrm{Hom}(E_1, E_2).$$

in r variables, where r is the rank of $\mathrm{Hom}(E_1, E_2)$.

The quadratic forms q_A and q_{E_1,E_2}

Let $A = E_1 \times E_2$. The relation between the intersection form and the degree map is

$$q_A(x, y, \varphi) = xy - q_{E_1,E_2}(\varphi). \tag{1}$$

Note that q_A is an indefinite integral quadratic form in $\rho = r + 2$ variables.

Refined Humbert Invariants

Suppose that A has a principal polarization $\theta \in \mathcal{P}(A)$. Then the quadratic form $\tilde{q}_{(A,\theta)}$ on $\mathsf{NS}(A)$ is

$$\tilde{q}_{(A,\theta)}(D) = (D \cdot \theta)^2 - 2(D \cdot D), \text{ for } D \in \mathsf{NS}(A). \tag{2}$$

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- The form $\tilde{q}_{(A,\theta)}$ leads to a quadratic form $q_{(A,\theta)}$ on the quotient module *polarized Neron Severi group* $\mathrm{NS}(A)/\mathbb{Z}\theta$.

Definition

A *refined Humbert invariant* $q_{(A,\theta)}$ of (A, θ) is a positive-definite integral quadratic form on $\mathrm{NS}(A, \theta)$ satisfying, for all $\mathrm{cl}(D) \in \mathrm{NS}(A, \theta)$

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Table 1: Dictionary of the notation

	$\mathrm{NS}(A)$	$\mathrm{NS}(A,\theta)$	$\mathrm{Hom}(E_1, E_2)$
Quadratic form	q_A	$q_{(A,\theta)}$	q_{E_1,E_2}
# variables	6	5	4
# variables	4	3	2

Equivalent definition

Definition
The endomorphisms which are symmetric with respect to the Rosati involution r_θ defined by θ

$$\mathrm{End}_\theta(A) = \{\mu \in \mathrm{End}(A) : \hat{\mu} \circ \theta = \theta \circ \mu\} = \{\mu \in \mathrm{End}(A) : \mu = \mu'\}$$

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Definition (Kani 2019)
For every $\mu \in \mathrm{End}_\theta(A)$, we define

$$q_{(A,\theta)}(\mu) = \mathrm{tr}\left(\mu^2\right) - \frac{1}{4}(\mathrm{tr}(\mu))^2.$$

Theorem (Kani, 2016)

Let $A = E_1 \times E_2$ be a product of two elliptic curves. We have a group isomorphism

$$\begin{aligned}\mathcal{D} = \mathcal{D}_{E_1,E_2} :& \mathbb{Z} \oplus \mathbb{Z} \oplus \mathrm{Hom}(E_1, E_2) \to \mathrm{NS}(A) \\ & (a, b, \varphi) \mapsto (a-1)\theta_1 + (b - \deg(\varphi))\theta_2 + \Gamma_{-\varphi}.\end{aligned}$$

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For two divisors $D_1 = \mathcal{D}(a, b, \varphi)$ and $D_2 = \mathcal{D}(a^{'}, b^{'}, \varphi^{'})$ in $\mathrm{NS}(A)$, the intersection number of the divisors is given by

$$(D_1 \cdot D_2) = ab^{'} + a^{'}b - \beta_d(\varphi, \varphi^{'}),$$

where β_d is the bilinear form associated to the degree map q_{E_1,E_2} given by

$$\beta_d(\varphi, \varphi') = q_{E_1,E_2}(\varphi + \varphi') - q_{E_1,E_2}(\varphi) - q_{E_1,E_2}(\varphi').$$

Theorem (Kani, 2016)

Let $A = E_1 \times E_2$*, and* $P(A)$ *be a set of principal polarizations. Let* $D = \mathcal{D}(a, b, h) \in \mathrm{NS}(A)$*. Then* $D \in P(A)$ *if and only if* $a > 0$ *and* $ab - \deg(h) = 1$*.*

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We can represent a divisor $D = \mathcal{D}(a, b, \varphi)$ by two integers and one isogeny φ. This gives us a way easier method of computation of the intersection numbers.

Refined Humbert invariants

- Curves of genus 2 curves which lie on a (CM, supersingular) product surface $A = E_1 \times E_2$ where $\mathrm{Hom}(E_1, E_2) \neq \emptyset$ and $\mathrm{End}(E_i) \neq \mathbb{Z}$ for $i = 1, 2$.

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- Let $A = E_1 \times E_2$ such that $\mathrm{End}(E_i) = \mathcal{O}$ for a maximal order in a quaternion algebra $\mathrm{B}_{p,\infty}$ ramified at p and ∞.

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- The refined Humbert invariant attached to a curve on such a product surface is a *quintic form*.

Polarizations

Definition
A polarization $\theta \in \mathcal{P}(A)$ is called *reducible* (or *decomposable*) if $\theta = \mathrm{cl}(E_1 + E_2)$ for some elliptic curves E_1 and E_2 on A. Otherwise, it is called *irreducible*.

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- Let θ be irreducible, then $D = C$ is a smooth curve of genus 2 on A such that $(A, \theta) \cong (\mathcal{J}(\mathcal{C}), \theta)$, where $\theta = cl(D)$.
- We use the notation $q_{\mathcal{C}}$ for the refined Humbert invariant $q_{(\mathcal{J}(\mathcal{C}),\theta)}$.

The irreducibility criterion

Theorem

Let $q_{(A,\theta)}$ be the refined Humbert invariant of the principally polarized abelian surface (A, θ), then we have that

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- The advantage of a refined Humbert invariants $q_{(A,\theta)}$ of a principally polarized abelian surface (A, θ) is the use of the intersection theory of divisors on surfaces.

Splitting

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Theorem
A curve $\mathcal{C}/K$ has an elliptic subcover of degree N if and only if $q_{\mathcal{C}}$ primitively represents N^2. Equivalently, $\mathcal{J}(\mathcal{C})$ is (N, N) split if and only if $q_{\mathcal{C}}$ primitively represents N^2.

Automorphism groups

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Theorem (Kani,2023)

If $\mathcal{C}/K$ is a curve of genus 2 , then

$$|i(\mathrm{Aut}(\mathcal{C})) - 1| = r_4(q_{\mathcal{C}}).$$

Thus, if $r_4(q_{\mathcal{C}}) > 0$, then $\mathcal{J}(\mathcal{C}) \sim E_1 \times E_2$ for some elliptic curves E_i/K, $i = 1, 2$. Moreover, if $r_4(q_{\mathcal{C}}) \geq 3$, then $E_1 \sim E_2$.

Quintic refined Humbert invariants

Lemma (Kani, 2019)

The determinant of the Néron-Severi group of $E_1 \times E_2$ with respect to the intersection form is given by

$$\det(\mathrm{NS}(E_1 \times E_2)) = (-1)^{\rho-1} \det(\mathrm{Hom}(E_1, E_2), \beta_d),$$

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Corollary (Kani, 2019)

If $\mathcal{C}/K$ has an elliptic subcover $F : \mathcal{C} \to E$ with $\mathrm{char}(K) \nmid \deg(f)$, *if the Picard number* $\rho = \mathrm{rank}(\mathrm{NS}(\mathcal{J}(\mathcal{C})) \geq 3$, *and if* $\det(q_{\mathcal{C}})/2^{2\rho-3}$ *is square-free, then $\mathcal{J}(\mathcal{C}) \cong E \times E'$ for some elliptic curve E'/K.*

Quintic refined Humbert invariants

Theorem (Kani 2019)

If $\mathcal{C}/K$ has a presentation (E, E', ψ) of degree N with $\mathrm{char}(K) \nmid N$ and isogeny defect $m = m_\psi$, and if $r = \mathrm{rank}(\mathrm{Hom}(E, E')) \geq 1$, then the refined Humbert invariant $q_{\mathcal{C}}$ is a positive definite quadratic form of rank $r+1$ which satisfies properties

(i) $\det(q_{\mathcal{C}}) = 2^{2r+1} m^2 \det\big(q_{E,E'}\big)$.

(ii) $q_{\mathcal{C}}$ *primitively represents* N^2.

(iii) $q_{\mathcal{C}}(x_1, \ldots, x_{r+1}) \equiv 0, 1 \pmod 4$ *for all* $x_1, \ldots, x_{r+1} \in \mathbb{Z}$.

(iv) $q_{\mathcal{C}}(x_1, \ldots, x_{r+1}) \neq 1$ *for all* $x_1, \ldots, x_{r+1} \in \mathbb{Z}$.

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Theorem (Kani, 2019)

If (E, E', ψ) is a presentation of degree N of a curve $\mathcal{C}/K$ of genus 2 with $\mathrm{char}(K) \nmid N$, then $m_\psi = 1$ if and only if $\mathcal{J}(\mathcal{C}) \cong E \times E'$.

Applications to isogeny-based cryptography

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- A security reduction between Computational Refined Humbert Invariant Problem and Isogeny Problem (for SQIsign and CSIDH)
- Isogeny degree analysis
- Testing the folklore, detecting (N, N) splittings and minimum walks

Security Reductions

Theorem (K., Martindale)

Computing the quintic refined Humbert invariant of $E \times E'$ in the context of SQISign is computationally equivalent up to polynomial-time algorithms to solving the isogeny problem.

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Theorem (K., Martindale)

Computing the ternary refined Humbert invariant of $E \times E'$ is computationally equivalent up to polynomial-time algorithms to solving the CSIDH isogeny problem.

Isogeny degree analysis

Lemma

Let E_1 and E_2 be elliptic curves over a field K, and let $A = E_1 \times E_2$ be the product surface with the product polarization $\theta_A = \theta_{E_1} \otimes \theta_{E_2}$. For $a, b \in \mathbb{Z}$, and $\varphi \in \mathrm{Hom}(E_1, E_2)$, then

$$q_{(A,\theta_A)}(\mathcal{D}(a, b, \varphi)) = (a - b)^2 + 4q_{E_1,E_2}(\varphi),$$

q_{E_1,E_2} denotes the degree form on $\mathrm{Hom}(E_1, E_2)$.

Computation of refined Humbert invariants

- *Refined Humbert invariants in supersingular isogeny degree analysis*
  `https://eprint.iacr.org/2025/1605` with G.Korpal
- Let $A = E \times E$ be a product surface for a supersingular curve E.
- Let $\mathcal{O}$ be a maximal quaternion order in $\mathrm{B}_{\mathrm{p},\infty}$ such that $\mathrm{End}(E) = \mathcal{O}$
- The corresponding $q_{(A,\theta)}$ quintic (quinary) integral quadratic form.
- Let θ be a reducible principal polarization (product polarization).
- We represent $\mathcal{D}(a, b, \varphi)$ as $\begin{pmatrix} a & \varphi \\ -\hat{\varphi} & b \end{pmatrix}$.

Isogeny degrees

- We need the understanding of the value over a fixed finite field $\mathbb{F}_{p^2}$

  $$\max_{q_{E_1,E_2}} \{\min\{N : q_{E_1,E_2}(x_1, \ldots, x_4) = N \text{ for some } x_1, \ldots, x_4 \in \mathbb{Z}\}\}.$$

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  This is only ranging over E_1 and E_2 supersingular and defined over $\mathbb{F}_{p^2}$.
- For this, we compute all the refined Humbert invariants over a fixed finite field. We pull out the degree maps from refined Humbert invariants, and compute this number.

$p = 107$, 581 quadratic forms

- Let $A = E \times E$ be surface, θ principal polarization.
- $p \equiv 11 \pmod{12}$ and $\mathcal{O} = \langle 1, \mathrm{i}, (\mathrm{i}+\mathrm{j})/2, (1+\mathrm{k})/2 \rangle$.
- Total polarizations checked: 63050.
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- $\theta = \begin{bmatrix} 12 & 12 - 11\beta_1 - 2\beta_2 - 2\beta_3 \\ 12 + 11\beta_1 + 2\beta_2 + 2\beta_3 & 40 \end{bmatrix}$,
- $q_{(A,\theta)} = x_1^2 + 8x_2^2 + 8x_2x_3 + 4x_2x_5 + 24x_3^2 - 20x_3x_4 - 8x_3x_5 + 24x_4^2 + 24x_4x_5 + 60x_5^2$.
- $q_{E_1,E_2} = 2x_1^2 + 2x_1x_2 + x_1x_4 + 6x_2^2 - 5x_2x_3 - 2x_2x_4 + 6x_3^2 + 6x_3x_4 + 15x_4^2$.

Data on isogeny degrees

We get the following results for allRHI(p) for primes $p \equiv 11 \pmod{12}$ with $10 < p < 420$. Here, d is the maximum of all minimum degrees. We observe that the minimum isogeny degree d has a complexity $\approx 0.62\sqrt{p}$.

p	$\#\Theta$	$\mathbf{H}$	$\#\mathrm{RHI}_{\mathrm{red}}$	$\#\mathrm{RHI}_{\mathrm{iso}}$	$\#\mathrm{Gen}(q_5)$ rep. 1	$\frac{\mathbf{h}(\mathbf{h}+1)}{2}$	d
11	$1331 = 11^3$	5	546	3	3	3	2
23	$12167 = 23^3$	16	2551	6	6	6	3
47	$103823 = 47^3$	72	10822	15	15	15	4
59	$81490 \approx 59^3/3$	125	6821	21	21	21	5
71	$144760 \approx 71^3/2$	198	9982	28	28	28	5
83	$138483 \approx 83^3/4$	296	8221	29	29	36	6
107	$63050 \approx 107^3/19$	581	2939	39	39	55	6
131	$105348 \approx 131^3/21$	1008	4052	67	67	78	7
167	$104102 \approx 167^3/45$	1978	3237	94	94	120	8
179	$104371 \approx 179^3/55$	2404	2947	97	97	136	9
191	$105386 \approx 191^3/66$	2886	2831	123	123	153	9
227	$105230 \approx 227^3/111$	4712	2324	133	135	210	10
239	$101640 \approx 239^3/134$	5460	2171	175	177	231	10
251	$101981 \approx 251^3/155$	6281	2097	179	181	253	10
263	$99055 \approx 263^3/184$	7182	1854	184	186	276	11
311	$87345 \approx 311^3/344$	11644	1346	274	286	378	11
347	$83564 \approx 347^3/500$	15993	1227	253	265	465	11
359	$86476 \approx 359^3/554$	17654	1173	316	346	496	12
383	$85104 \approx 383^3/660$	21310	1069	322	361	561	12
419	$85323 \approx 419^3/862$	27692	1001	359	423	666	12

Split surfaces and minimum walks

Testing the folklore, detecting split surfaces, and minimum walks on isogeny graphs of superspecial abelian surfaces joint work with G.Korpal

- $p \equiv 11 \pmod{12}$ and $\mathcal{O} = \langle 1, \mathrm{i}, (\mathrm{i}+\mathrm{j})/2, (1+\mathrm{k})/2 \rangle$.
- Let $A = E \times E$ be a superspecial product surface.
- Choose an irreducible principal polarization θ.
- For every positive integer N^2 primitively represented by $q_{\mathcal{C}}$, there exists an (N, N)-isogeny from $\mathcal{J}(\mathcal{C})$ to a split abelian surface $E_1 \times E_2$.

Split surfaces and minimum walks

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Let $N = 2^k$. An algorithm of primitive representation of 4^k given refined Humbert invariants $q_{\mathcal{C}}$ gives us find splittings of $(2^k, 2^k)$-isogenies in the Richelot isogeny graph.

Example

Table 2: The splitting of surfaces for $p = 167$ by $(2^k, 2^k)$-isogenies.

RHI type	primitive solution for $1 \leq k \leq 31$	# no solution $1 \leq k \leq 31$
1	$1, \ldots, 22, 24, 27, 29, 31$	none
2	$2, \ldots, 31$	1
3	$1, \ldots, 31$	none
4	$2, \ldots, 31$	1
5	$1, 3, \ldots, 21, 25, 26, 27, 30$	2
6	$2, \ldots, 31$	1
7	$1, \ldots, 31$	none
8	$2, \ldots, 31$	1
9	$2, \ldots, 24, 28, 29, 30, 31$	1
10	$2, \ldots, 31$	1

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- Minimum distance to split surfaces among the 107 samples:
  - 2 RHI cases: 1 step distance,
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  For $18\log(1103) + 9 \leq k \leq 10000$:
  - 65 invariants were highly primitively representable by 4^k,
  - 42 failed primitive representability and local conditions.

Automorphism groups

Table 3: Data summary. Comparing Theorem IKO with RHI for random polarizations.

p	$\#\Theta$	$\#\mathrm{RHI}_{\text{irred}}$	$\#\mathrm{RHI}_{\text{iso}}$	$\#\mathrm{Gen}\,q_5$ not 1	$\#\,\mathrm{Aut}(\mathcal{C})$ type: (`Thm IKO count`, `RHI count`)						
					C_2	C_{10}	$\mathrm{C}_2 \times \mathrm{C}_2$	D_4	D_6	$\mathrm{C}_3 \rtimes \mathrm{D}_4$	$\mathrm{GL}_2(\mathbb{F}_3)$
11	$1331 = 11^3$	785	2	2	$(0,0)$	$(0,0)$	$(0,0)$	$(0,0)$	$(1,1)$	$(1,1)$	$(0,0)$
23	$12167 = 23^3$	9616	9	9	$(0,0)$	$(0,0)$	$(5,4)$	$(1,1)$	$(2,2)$	$(1,1)$	$(1,1)$
47	$103823 = 47^3$	93001	43	43	$(14,9)$	$(0,0)$	$(31,23)$	$(4,4)$	$(6,5)$	$(1,1)$	$(1,1)$
59	$81490 \approx \frac{59^3}{3}$	74669	74	74	$(35,25)$	$(1,1^*)$	$(52,35)$	$(6,5)$	$(9,7)$	$(1,1)$	$(0,0)$
71	$144760 \approx \frac{71^3}{2}$	134778	116	116	$(70,42)$	$(0,0)$	$(81,58)$	$(7,6)$	$(10,8)$	$(1,1)$	$(1,1)$
83	$138483 \approx \frac{83^3}{4}$	130262	165	165	$(123,80)$	$(0,0)$	$(114,67)$	$(9,7)$	$(13,10)$	$(1,1)$	$(0,0)$
107	$63050 \approx \frac{107^3}{19}$	60111	311	331	$(296,178)$	$(0,0)$	$(200,111)$	$(12,8)$	$(17,13)$	$(1,1)$	$(0,0)$
131	$105348 \approx \frac{131^3}{21}$	101296	554	554	$(583,340)$	$(0,0)$	$(310,186)$	$(15,11)$	$(21,16)$	$(1,1)$	$(0,0)$
167	$104102 \approx \frac{167^3}{45}$	100865	1058	1058	$(1290,709)$	$(0,0)$	$(521,313)$	$(19,15)$	$(26,19)$	$(1,1)$	$(1,1)$
179	$104371 \approx \frac{179^3}{55}$	101424	1266	1266	$(1614,894)$	$(1,1^*)$	$(602,336)$	$(21,14)$	$(29,20)$	$(1,1)$	$(0,0)$
191	$105386 \approx \frac{191^3}{66}$	102555	1540	1540	$(1988,1079)$	$(0,0)$	$(691,420)$	$(22,17)$	$(30,22)$	$(1,1)$	$(1,1)$
227	$105230 \approx \frac{227^3}{111}$	102906	2425	2425	$(3447,1848)$	$(0,0)$	$(990,533)$	$(27,18)$	$(37,25)$	$(1,1)$	$(0,0)$
239	$101640 \approx \frac{239^3}{134}$	99469	2868	2869	$(4057,2164)$	$(1,1^*)$	$(1103,654)$	$(28,21)$	$(38,27)$	$(1,1)$	$(1,0)$
251	$101981 \approx \frac{251^3}{155}$	99884	3269	3271	$(4736,2541)$	$(0,0)$	$(1220,680)$	$(30,19)$	$(41,28)$	$(1,1)$	$(0,0)$
263	$99055 \approx \frac{263^3}{184}$	97201	3704	3709	$(5486,2904)$	$(0,0)$	$(1345,751)$	$(31,22)$	$(42,27)$	$(1,0)$	$(1,0)$

Thanks for listening!

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